Noise in Constrained QAOA on IBM Heron Hardware: Landscape Compression Tracks Feasibility Loss

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***Abstract*—We study how hardware noise changes the p=1 Quantum Approximate Optimization Algorithm (QAOA) landscape for cardinality-constrained portfolio-selection problems on *ibm_fez* (IBM Heron r2), across seven instances from n=6 to n=16 qubits, each measured on a 7×7 parameter grid at both 4,096 and 8,192 shots per point. The first version of this paper proposed Landscape Span Compression (LSC), the fractional loss of energy range across the grid, as a device benchmark, based on stable values of 0.24–0.30 at $n\leq8$. The larger study shows why that does not hold. Hardware LSC rises to 0.72–0.81 at n=16 and is well described by an exponential decay in transpiled CZ count (R^2=0.96 and 0.88 at the two shot counts). A statistic built only from the fraction of feasible shots reproduces it (Pearson r=0.954, p=0.0009), and the energy of feasible shots alone is unchanged by noise to within 0.02. For penalty-encoded constrained problems, landscape compression is therefore a circuit-fidelity effect expressed as loss of feasible samples, as noise-flattening models of QAOA predict, not a change in landscape shape. The same mechanism affects the usual test of classical-to-hardware parameter transfer: grid verdicts on whether the optimum moves agree across shot counts for only 2 of 7 instances, and where the hardware optimum does move, it moves toward higher feasibility. Best-feasible approximation ratios are $\geq$ 0.9993 throughout, but the feasible sets are nearly degenerate in energy, so this is a weak test of solution quality on these instances. We withdraw the benchmark and noise-model-gap claims of the first version, correct its solution-quality figures, and describe a measurement protocol that separates feasibility loss from solution-quality effects.**

***Index Terms*—QAOA, noise characterization, constrained optimization, NISQ, parameter transfer, metric validity, reproducibility**

I. Introduction

The Quantum Approximate Optimization Algorithm (QAOA) [1] is the most widely run variational algorithm for combinatorial optimization on noisy intermediate-scale hardware [2]. Hardware studies have measured how noise degrades its approximation ratio on unconstrained problems [6], [7], and barren-plateau results bound how trainable it remains as circuits grow [5]. In practice its two angles per layer are usually optimized in classical simulation and then executed on the device, which saves scarce QPU time but assumes that noise does not move the optimum. Analytical and numerical work supports that assumption for simple noise models: local unital noise scales the cost landscape toward its maximally mixed value without changing its shape [14], [15], and optimal parameters are resilient to several noise types [17]; symmetry arguments further support transferring parameters between instances [8]. The same models imply that the landscape flattens exponentially with circuit size, which makes the optimum harder to locate [16].

Constrained problems add a complication that these models treat only implicitly. Constraints can be enforced by constraint-preserving mixers [12] or warm starts [13], but are most often encoded as a penalty. When a constraint is encoded as a penalty in a Quadratic Unconstrained Binary Optimization (QUBO) objective, most bitstrings are infeasible and carry large penalty energy. Noise that pushes the output toward uniform sampling then changes the measured energy mainly by changing the fraction of feasible samples, not by changing the energies of feasible solutions.

The first version of this paper [18] introduced Landscape Span Compression (LSC), the fractional loss of energy range over a parameter grid, and proposed it as a device benchmark after observing stable hardware values of 0.24–0.30 for three instances at $n \in \{6, 8\}$ on *ibm_fez*. It also reported that the classical optimum transferred to hardware in every case (OPS=0). This version reports a scaling study to n=16 under a NERSC QCAN allocation, with every instance measured at two shot counts. Its findings replace the first version's central claims:

- Hardware LSC grows with circuit size and is explained by two-qubit gate count; it is not a device-level quantity (Section III-A).
- The growth is almost entirely the loss of feasible samples, consistent with the noise-flattening model applied to a penalty-encoded QUBO (Sections III-B and III-C).
- Grid-based tests of parameter transfer are not reproducible at this resolution and are themselves biased toward feasibility (Section III-D).
- Best-feasible approximation ratios near 1 are close to automatic on these instances and do not by themselves

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measure QAOA's effect (Section III-E).

Section V lists every claim of the first version and its status. We report these results as an experience study because the way a small-n characterization fails at larger n is directly useful to practitioners who characterize devices or transfer parameters from simulation.

II. Problem and Methods

A. Constrained QUBO instances

Each instance selects $k{=}n/2$ of n assets:

$$\min_{\mathbf{x}\in\{0,1\}^n} \; -\mu^\top \mathbf{x} + \lambda\, \mathbf{x}^\top \Sigma\, \mathbf{x} + P\Big(\sum_i x_i - k\Big)^2, \qquad (1)$$

with mean returns μ and covariance Σ estimated from daily prices (January 2019 to December 2023, `yfinance`), and penalty $P = \|\mu\|_1 + \lambda\|\sigma\|_1$ chosen to dominate the objective [3], [4]. The resulting QUBO matrix Q defines the cost Hamiltonian. Table I lists the instances. The feasible set has $\binom{n}{k}$ members, a fraction 31%, 27%, 23% and 20% of all bitstrings for $n{=}6$, 8, 12 and 16.

TABLE I
INSTANCES. LOW-VOL AND HIGH-VOL USE LOWER- AND HIGHER-VOLATILITY TICKER SETS; 8 TSLA IS THE HIGH-VOLATILITY $n{=}8$ INSTANCE OF THE FIRST VERSION.

Instance	n	$\binom{n}{k}$	Tickers
6 low-vol	6	20	AAPL MSFT JPM XOM JNJ AMZN
8 low-vol	8	70	6 low-vol + GOOGL BRK-B
8 TSLA	8	70	6 low-vol + TSLA IBM
12 low-vol	12	924	8 low-vol + PG KO WMT MRK
12 high-vol	12	924	6 low-vol + TSLA IBM NVDA META COIN PLTR
16 low-vol	16	12,870	12 low-vol + PEP UNH HD MCD
16 high-vol	16	12,870	12 high-vol + AMD TSM XYZ RIVN

B. Circuits, grid and execution

We run $p{=}1$ QAOA from $|+\rangle^{\otimes n}$ with cost unitary $e^{-i\gamma H_C}$ and transverse-field mixer $e^{-i\beta\sum_i X_i}$. A reference point (γ^*, β^*) is found by multi-start COBYLA on the noiseless simulator and then fixed. The landscape is sampled on a $7{\times}7$ grid centred on (γ^*, β^*) with half-width 0.4 rad (spacing 0.133 rad) under three conditions: noiseless simulation, simulation with the calibration noise model (`AerSimulator.from_backend`), and hardware. Hardware runs used *ibm_fez* (Heron r2, 156 qubits) through `SamplerV2` in batch mode, with circuits transpiled to $\{CZ, R_Z, SX, X\}$ at optimization level 3. Every instance was run at both 4,096 and 8,192 shots per grid point with the same fixed (γ^*, β^*). The transpiled circuits contain about 48, 103, 250 and 500 CZ gates at $n{=}6$, 8, 12 and 16.

C. Quantities

At each grid point the sampler returns counts $\{x_j \mapsto n_j\}$. The landscape value is the shot-averaged QUBO cost

$$L(\gamma,\beta) = \sum_j \frac{n_j}{N_{\text{shots}}}\, x_j^\top Q\, x_j, \qquad (2)$$

averaged over *all* shots, feasible or not. The ideal condition is also shot-sampled. From L we compute:

- **Landscape span** $\mathrm{LS}(L) = \max L - \min L$ over the grid, and **Landscape Span Compression** $\mathrm{LSC} = 1 - \mathrm{LS}(L_{\text{cond}})/\mathrm{LS}(L_{\text{ideal}})$.
- **Feasible fraction** $\mathrm{FF}(\gamma,\beta)$, the fraction of shots with $\sum_i x_i = k$.
- **Optimal-parameter shift** (OPS): whether the grid cell minimising L_{hw} equals the cell minimising L_{ideal}.
- **Best-feasible approximation ratio** $\mathrm{AR} = E_{\text{bf}}/E^*$, where E_{bf} is the lowest cost among feasible shots at a condition's best grid cell and E^* is the brute-force optimum. The first version instead used the shot-averaged L in the numerator.

III. Results

A. LSC grows with circuit size

Table II gives LSC for all seven instances. Hardware LSC rises from 0.23–0.32 at $n{\le}8$ to 0.72–0.81 at $n{=}16$ at both shot counts, and the calibration noise model shows the same trend at lower values. Fitting $1-\mathrm{LSC}_{\text{hw}} = e^{-kN_{CZ}}$ across the seven instances gives $k = 0.00298 \pm 0.00028$ per CZ ($R^2{=}0.959$) at 4,096 shots and $k = 0.00261 \pm 0.00044$ ($R^2{=}0.876$) at 8,192 shots (Fig. 1). The metric also saturates: the $n{=}16$ points barely exceed the $n{=}12$ points despite about twice the CZ count, as expected for a quantity bounded by 1. The stability seen at $n{\le}8$ reflected a narrow range of circuit sizes.

TABLE II
LSC ON *ibm_fez*, $7{\times}7$ GRID. "N": CALIBRATION NOISE MODEL; "HW": HARDWARE.

	4,096 shots		8,192 shots	
Instance	$\mathrm{LSC_n}$	$\mathrm{LSC_{hw}}$	$\mathrm{LSC_n}$	$\mathrm{LSC_{hw}}$
6 low-vol	0.155	0.248	0.135	0.244
8 low-vol	0.181	0.240	0.163	0.228
8 TSLA	0.197	0.291	0.181	0.322
12 low-vol	0.319	0.550	0.332	0.585
12 high-vol	0.425	0.660	0.431	0.651
16 low-vol	0.419	0.811	0.479	0.716
16 high-vol	0.573	0.784	0.597	0.812

B. The compression is loss of feasible samples

Because L averages over all shots, it mixes two things: which bitstrings are feasible and how good the feasible ones are. The penalty term separates them sharply in energy. For 12 low-vol, all 924 feasible bitstrings lie within 0.489 of the optimum $E^*{=}-156.36$, while a bitstring one asset away from feasibility averages about -151.8, nine times the entire

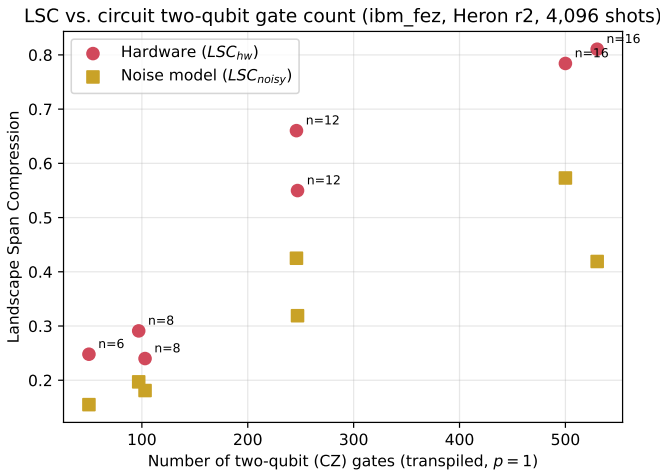


Fig. 1. $\mathrm{LSC_{hw}}$ and $\mathrm{LSC_n}$ against transpiled CZ count, seven instances on *ibm_fez*, 4,096 shots per point.

feasible range further away. Fig. 2 shows the resulting shell structure of the sampled energies.

Splitting L at the grid centre makes the effect explicit (Table III). The all-shots energy rises by 3.65 from ideal to hardware, while the energy of feasible shots alone changes by 0.02. All of the apparent compression at this point is the feasible fraction falling from 37.1% to 28.5%.

TABLE III
12 LOW-VOL AT THE GRID CENTRE, 4,096 SHOTS.

Condition	FF	L (all shots)	L (feasible only)
Ideal	37.1%	−151.21	−156.14
Noisy	32.6%	−149.52	−156.14
Hardware	28.5%	−147.56	−156.12

The same holds across the grid and across instances. We define a feasibility-range compression that mirrors LSC but uses no energies, $1 - \mathrm{range}(\mathrm{FF_{hw}})/\mathrm{range}(\mathrm{FF_{ideal}})$. It tracks hardware LSC across all seven instances with Pearson r=0.954 (p=0.0009; Fig. 3). The mean feasible fraction over the grid barely changes (for 12 low-vol, 0.201, 0.200 and 0.202 for ideal, noisy and hardware); what collapses is its spread. In the ideal landscape, a sharp feasibility peak near the optimum is what gives L its range, and noise erases it (Fig. 4).

C. Why this is expected

Under local unital noise, the noisy QAOA cost is well approximated by [14], [15]

$$L_\varepsilon \approx F\,L_0 + (1-F)\,\bar{C}, \qquad F \approx (1-\varepsilon)^N, \tag{3}$$

where L_0 is the noiseless landscape, $\bar{C}$ the cost averaged over all bitstrings, and F a survival factor over N noisy operations. Because the second term is constant across the grid, $\mathrm{LS}(L_\varepsilon) \approx F\,\mathrm{LS}(L_0)$ and $\mathrm{LSC} \approx 1-F$. LSC is thus an estimate of circuit fidelity, which decays exponentially in gate count, as observed. For a penalty-encoded problem, $\bar{C}$ is dominated by infeasible bitstrings, so the loss of fidelity appears as a loss of feasible samples while the feasible energies themselves stay put, as in Table III. The fitted k values correspond to a per-CZ decay of 0.26–0.30%, the same order as two-qubit error rates reported for Heron devices; we do not claim a closer match, because k also absorbs single-qubit, idle and readout error. The model assumes unital noise and ignores amplitude damping, which on superconducting hardware relaxes toward $|0\cdots0\rangle$ rather than the uniform state.

Equation (3) also explains why the optimum becomes hard to locate. Near the optimum $L_\varepsilon \approx L_{\min} + \frac{1}{2}F\kappa\,\delta^2$, where κ is the noiseless curvature. With per-point statistical error σ, displacements smaller than about $\sqrt{2\sigma/(F\kappa)}$ cannot be resolved, and this limit grows as F falls with circuit size [16].

D. Parameter transfer

Table IV reports OPS at both shot counts. Only 2 of 7 instances give the same verdict; 4 flip, and 16 high-vol is violated at both but at different cells on opposite sides. At a grid spacing of 0.133 rad, individual verdicts are not reproducible.

TABLE IV
OPS ON *ibm_fez*: IDEAL AND HARDWARE ARGMIN CELLS (ROW, COLUMN) ON THE 7×7 GRID.

		4,096 sh.		8,192 sh.	
Instance	**Ideal**	**HW**	**OPS**	**HW**	**OPS**
6 low-vol	(3,3)	(3,4)	viol.	(3,3)	holds
8 low-vol	(3,3)	(3,3)	holds	(3,3)	holds
8 TSLA	(3,3)	(3,3)	holds	(3,3)	holds
12 low-vol	(4,0)	(3,3)	viol.	(4,0)	holds
12 high-vol	(3,2)	(3,2)	holds	(3,1)	viol.
16 low-vol	(3,2)	(3,1)	viol.	(3,2)	holds
16 high-vol	(3,3)	(3,2)	viol.	(3,4)	viol.

Two further checks, using no additional QPU time, sharpen this. First, the reference point itself was not converged at larger n, because COBYLA was run on a 4,096-shot estimator. Re-optimizing each reference point on the exact statevector expectation (8 starts, Nelder–Mead) lowered the energy by 0.05–0.13 at $n \leq 8$ but by 2.0, 4.8, 10.1 and 4.4 for 12 low-vol, 12 high-vol, 16 low-vol and 16 high-vol. For 12 and 16 low-vol, the converged optimum lies in a different cell from the reported ideal argmin, and against it the two "holds" verdicts at 8,192 shots become violations. For 8 TSLA, 12 high-vol and 16 high-vol, the converged optimum lies outside the measured window, so OPS cannot be evaluated from the existing data.

Second, OPS inherits the feasibility bias of L. In all six violations, the hardware argmin has more feasible shots than the ideal argmin (21.9–47.6% against 20.0–45.8%): on hardware, the minimum of the all-shots energy tends to sit where the most shots are feasible. A grid OPS verdict computed on L therefore mixes a shift in the solution-quality optimum with the feasibility effect of Section III-B. Answering the transfer question requires measuring it on a quantity that separates the two (Section IV-B).

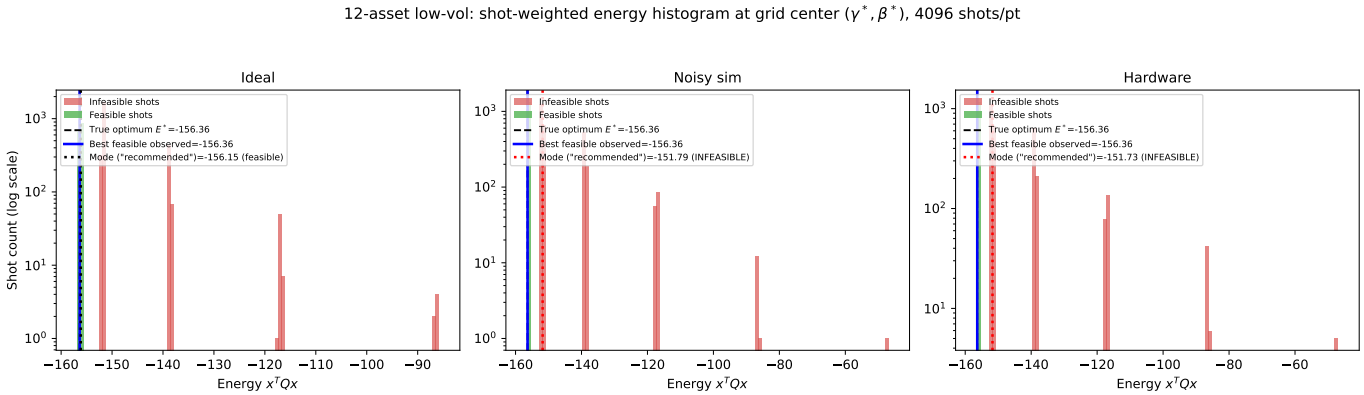


Fig. 2. Shot-weighted energy histogram at the grid centre for 12 low-vol (4,096 shots, log scale). Feasible shots form a narrow band at the optimum; infeasible shots fall in penalty shells. Noise moves weight from the feasible band into the shells. Under noise, the most frequent bitstring (dotted) is infeasible.

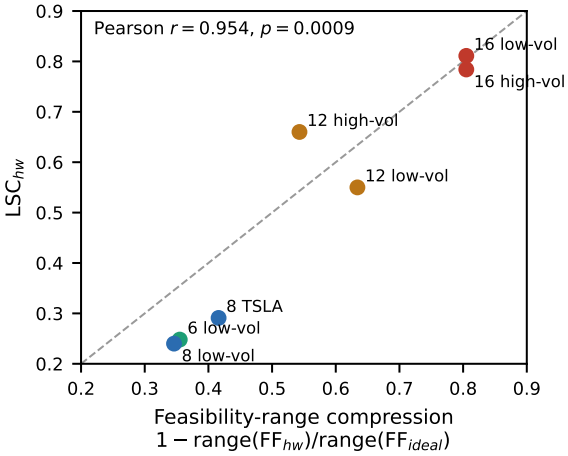


Fig. 3. Hardware LSC against feasibility-range compression, which uses only the feasible-shot fraction per grid point. Dashed: identity line.

E. Solution quality

Table V compares the approximation ratio as computed in the first version with the best-feasible ratio. The first version's figures (hardware 0.914–0.937, a "consistent noise cost of about 0.03") measured the average shot, not the best solution obtained. The best-feasible ratio is ≥ 0.9993 in all 42 instance, shot-count and condition combinations, including every hardware run. The first version also reported the most frequent bitstring as the recommended portfolio; that bitstring is infeasible in 11 of 42 combinations (26%), mostly under noise and at larger n.

This is a weak test on these instances. Their feasible sets are nearly degenerate in energy: every feasible solution is within 0.31% of the optimum for 12 low-vol and within 0.63–1.71% for $n \leq 8$, so any feasible sample scores AR ≥ 0.98 and most score above 0.99. Best-feasible AR near 1 mainly shows that feasible shots were produced. For the same reason, the solution-quality cost of the OPS violations above, which is at most 0.00013 in best-feasible AR, is close to guaranteed and says little about transfer.

TABLE V
APPROXIMATION RATIO RANGES OVER THREE CONDITIONS AND TWO SHOT COUNTS (6 COMBINATIONS PER INSTANCE). "MODE INF.": COMBINATIONS WHOSE MOST FREQUENT BITSTRING IS INFEASIBLE.

Instance	Shot-avg. (v1)	Best-feasible	Mode inf.
6 low-vol	0.905–0.945	1.0000–1.0000	0/6
8 low-vol	0.932–0.964	1.0000–1.0000	1/6
8 TSLA	0.940–0.965	1.0000–1.0000	0/6
12 low-vol	0.939–0.979	0.9999–1.0000	2/6
12 high-vol	0.937–0.974	0.9999–1.0000	2/6
16 low-vol	0.946–0.983	1.0000–1.0000	3/6
16 high-vol	0.951–0.980	0.9993–0.9999	3/6

A test that isolates QAOA's contribution compares the probability of sampling the exact optimum, $P(x^*)$, with the uniform-feasible baseline $\text{FF}/\binom{n}{k}$, and the mean energy of feasible shots with the mean over all feasible bitstrings. **[Pending:** $P(x^*)$ and feasible-conditional mean energy for all instances, conditions and argmin cells, from the stored counts; table and one paragraph.**]**

IV. Discussion

A. What LSC measures

For penalty-encoded constrained problems, LSC over the all-shots energy is a circuit-fidelity estimate. It says how much of the output has been replaced by near-uniform samples, which for these problems means infeasible ones. It carries little information about the shape of the landscape or the quality of feasible solutions, and it scales with circuit size, so it cannot be compared across circuits or devices without controlling for gate count. Established gate- and circuit-level benchmarks already measure fidelity directly. We therefore withdraw LSC as a device benchmark. As a diagnostic, it remains a cheap way to see feasibility collapse in a grid scan that has already been run.

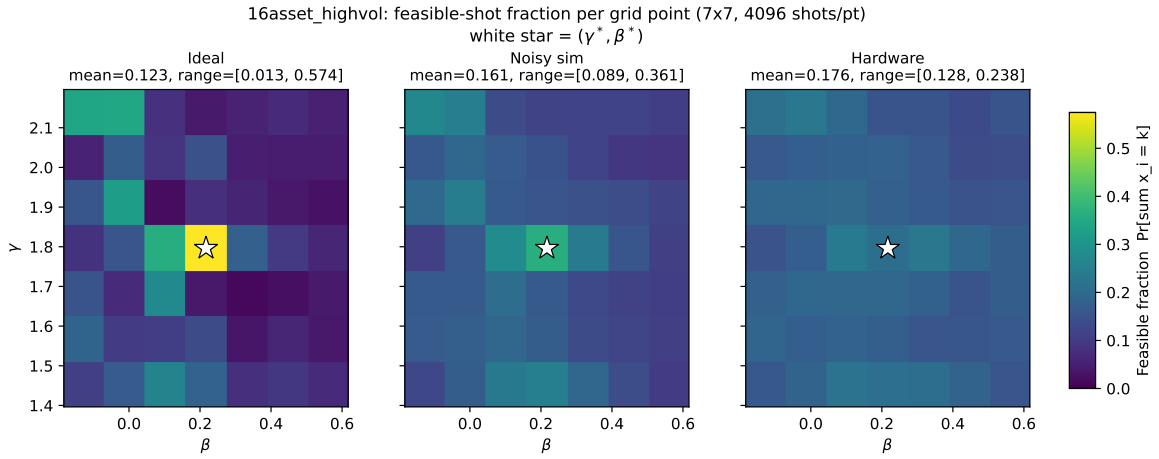


Fig. 4. Feasible-shot fraction per grid point, 16 high-vol (4,096 shots). The ideal landscape has a sharp feasibility peak (0.574) at (γ^*, β^*) (star); on hardware the range collapses to $[0.128, 0.238]$.

B. Measuring parameter transfer

The transfer question is still the practically important one, since it decides whether hardware-in-the-loop optimization can be skipped. Our results suggest four requirements for answering it on hardware:

1) *Converged references.* Fix (γ^*, β^*) on the exact expectation, not a shot-sampled estimator, and centre the search on it.
2) *A solution-quality target.* Locate the optimum of a quantity that separates QAOA's effect from feasibility loss, such as $P(x^*)$ relative to the uniform-feasible baseline, rather than the minimum of the all-shots energy.
3) *Resolution, not a grid cell.* Estimate the hardware optimum continuously by adaptive local search, with repeated same-point measurements in one session to separate shot noise from drift, and test displacements against the resolution limit $\sqrt{2\sigma/(F\kappa)}$.
4) *Fixed layouts.* Pin the physical qubit layout and repeat on a second patch, since placement was left to the transpiler here and chip-scale noise is not uniform.

Non-unital relaxation is the mechanism most likely to move the optimum genuinely, since it breaks the parameter independence of the fixed point in Eq. (3); a displacement that tracks T_1 relative to circuit duration more closely than two-qubit error would indicate it.

C. Lessons for practitioners

- A metric that is stable over a narrow range of circuit sizes can still be a circuit-size proxy. Test candidate device metrics over at least a factor of four in gate count.
- For constrained problems, report feasible fraction and feasible-conditional quantities separately from all-shots energy.
- Report the best feasible sample, never the most frequent bitstring, as the solution: the mode was infeasible in 26% of our combinations.
- Check that the instance's feasible set is not near-degenerate before using approximation ratio as evidence of solution quality.
- Do not trust a single grid argmin as evidence that parameters transfer; repeat it at a second shot count at minimum.

D. Limitations

All results are for p=1, one device (*ibm_fez*), one family of single-constraint portfolio problems, and an unpinned qubit layout. Grid mesh matters: refining the pilot scans from 7×7 to 13×13 raised hardware LSC by 0.037–0.039. The two shot-count runs of the same instance were taken up to a month apart, so their differences mix shot noise with calibration drift. The feasible sets of these instances are too degenerate for approximation ratio to discriminate solution quality.

V. Changes From Version 1

Table VI lists the claims of the first version [18] and their status.

The first version's data at $n\in\{6, 8\}$ included a 13×13 scan at 57,344 shots for 6 low-vol. This version uses 7×7 scans for all seven instances so that the comparison across n uses a common mesh.

VI. Conclusion

On IBM Heron hardware, noise in penalty-encoded constrained QAOA shows up almost entirely as loss of feasible samples. Landscape span compression measures that loss, grows with two-qubit gate count as circuit fidelity does, and is not a device benchmark. Grid tests of whether classically optimized parameters transfer to hardware are not reproducible at practical resolutions and inherit the same feasibility bias. Best-feasible approximation ratios near 1 are easy to obtain on near-degenerate instances and do not by themselves show that QAOA helps. The transfer question remains open and important; answering it requires converged references, a solution-quality target that is separated from feasibility, and continuous,

TABLE VI
CLAIMS OF VERSION 1 AND THEIR STATUS IN THIS VERSION.

Version 1 claim	**Status**
Hardware LSC of 0.24–0.30 is stable and can serve as a device benchmark	Withdrawn: grows with CZ count; tracks feasibility loss
OPS=0: classical parameters transfer directly to hardware	Not established: grid verdicts not reproducible, reference points unconverged at $n \geq 12$, verdicts biased by feasibility
Calibration noise model explains 44–59% of hardware LSC; Pearson r=0.959 structural agreement	Withdrawn: both computed on the all-shots energy
Consistent noise cost of ≈ 0.03 AR units	Withdrawn: measured the average shot; best-feasible AR ≥ 0.9993, but uninformative here
Feasible fraction 1.5–1.7× above random at (γ^*, β^*)	Unchanged for the pilot instances
ZNE [9]–[11]: $+7\%/+9\%/-4\%$ energy change, 3–5× uncertainty inflation	Unchanged as measured; applies to the all-shots energy, not to solution quality

repeated measurement of the hardware optimum, which is the subject of ongoing work.

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